Department of Electrical and Computer Engineering
North South University

Directed Research

Context Segmentation in Large Language Models: Towards Efficient Multi-Context Understanding in Prompt Processing

Talha Taiyoub **ID# 1921949042**

Madiha Jarrin **ID# 2022223642**

Faculty Advisor:

Dr. Mohammad Ashrafuzzaman Khan

Associate Professor

ECE Department

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LETTER OF TRANSMITTAL

May, 2025

To
Dr. Mohammad Abdul Matin
Chairman,
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: Submission of Research Report titled "Context Segmentation in Large Language Models: Towards Efficient Multi-Context Understanding in Prompt Processing"

Dear Sir,
With due respect, we, Talha Taiyoub and Madiha Jarrin, are pleased to submit our research report titled "Context Segmentation in Large Language Models: Towards Efficient Multi-Context Understanding in Prompt Processing," prepared as part of our academic requirements at North South University.

This report reflects our detailed study, analysis, and implementation of an advanced framework aimed at separating and understanding multiple contexts embedded within single prompts using large language models. We have also explored the potential of retrieval-augmented generation (RAG) to enhance accuracy and efficiency in context segmentation. The report includes background research, system design, methodology, implementation, analysis, results, limitations, and proposed future improvements.

We sincerely hope that the report meets your expectations and serves the purpose of academic evaluation. We would be grateful for your kind review and valuable feedback..

Sincerely,

Talha Taiyoub

Madiha Jarrin

ECE Department

North South University, Bangladesh.

APPROVAL

Talha Taiyoub (ID # 1921949042) and Madiha Jarrin (ID # 2022223642) from Electrical and Computer Engineering Department of North South University, have worked on the Directed Research Project titled "Context Segmentation in Large Language Models: Towards Efficient Multi-Context Understanding in Prompt Processing" under the supervision of Dr. Mohammad Ashrafuzzaman Khan partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

Supervisor's Signature

…….

Dr. Mohammad Ashrafuzzaman Khan

Associate Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

Chairman's Signature

…….

Dr. Mohammad Abdul Matin

Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

DECLARATION

This is to declare that this project/directed research is our original work. No part of this work has been submitted elsewhere, partially or fully, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names & Signatures

1. Talha Taiyoub

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2. Madiha Jarrin

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Furthermore, the authors would like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh, for facilitating the research. The authors would also like to thank their friends and family for their countless sacrifices and continual support.

ABSTRACT

Context Segmentation in Large Language Models: Towards Efficient Multi-Context Understanding in Prompt Processing

Large Language Models (LLMs) have shown impressive capabilities in natural language understanding, generation, and reasoning. However, their performance often deteriorates when processing prompts containing multiple intertwined contexts, resulting in conflated or incomplete responses. This research investigates the problem of context segmentation, the task of identifying and isolating distinct sub-contexts within a single prompt, to enable more accurate and structured language model outputs. We propose a framework that leverages Retrieval-Augmented Generation (RAG) to enrich the model's contextual awareness and improve segmentation accuracy. Our approach combines attention-based context modeling, prompt decomposition strategies, and post-hoc refinement techniques. We benchmarked this method against standard LLM behavior using both synthetic and real-world multi-context datasets. Quantitative evaluations demonstrate significant improvements in context separation accuracy, response relevance, and information completeness. Additionally, we explored practical applications of context segmentation, such as multi-image generation and speech-to-visual storytelling, to illustrate its potential for real-world deployment. This study lays the groundwork for more intelligent, adaptable, and context-aware prompt processing in advanced language systems.

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Chapter 1 Introduction

1.1 Background and Motivation

Large Language Models have revolutionized natural language processing by enabling systems to generate coherent, contextually aware responses to diverse prompts. Their ability to understand nuanced language and generate contextually rich content has found applications across education, customer service, content creation, and more. However, a recurring limitation arises when a single prompt contains multiple embedded contexts. In such cases, LLMs often struggle to maintain clarity, leading to partial, merged, or incorrect responses. This issue becomes more critical in applications like instruction following, conversational AI, and multimodal processing, where distinct topics must be addressed separately and accurately.

The motivation for this research stems from the growing need to enhance LLMs' interpretability and response fidelity in complex prompting scenarios. As users increasingly interact with LLMs using compound or multi-layered queries, effective context segmentation becomes essential. Traditional models tend to treat the prompt as a single monolithic input, ignoring the potential internal structure. This gap presents both a technical challenge and an opportunity for innovation.

By focusing on context segmentation, we aim to improve how language models process, interpret, and respond to multi-context prompts. Furthermore, with the rise of applications involving cross-modal generation, such as transforming speech into imagery, the importance of accurately separating contexts is amplified. This work seeks to address these challenges and open new directions for prompt understanding in intelligent systems.

1.2 Purpose and Goal of the Project

The primary purpose of this project is to explore and develop effective methods for identifying and separating multiple contexts within a single prompt, thereby enhancing the interpretive capabilities of Large Language Models. As LLMs are increasingly applied in real-world scenarios involving layered instructions, diverse intents, or complex narratives, the need to

process each context independently becomes crucial for generating accurate and meaningful responses.

The main goal is to design and evaluate a context segmentation framework that leverages Retrieval-Augmented Generation (RAG) to improve the model's ability to distinguish, isolate, and reformulate individual contexts from compound prompts. This framework aims to transform a dense or multi-faceted input into structured components that can be addressed with clarity and precision. In addition to theoretical development, the project also intends to demonstrate practical applications such as prompt-to-multiple-image generation and speech-to-visual storytelling, where context segmentation plays a pivotal role in aligning AI outputs with user expectations.

Ultimately, the project seeks to contribute to the broader field of prompt engineering by laying the foundation for context-aware systems that are more intelligent, adaptable, and aligned with human communication patterns.

1.3 Organization of the Report

This report is organized into five comprehensive chapters, each focusing on a distinct aspect of the research project.

Chapter 1: Introduction outlines the background, motivation, and objectives of the project, and presents the overall structure of the report.

Chapter 2: Research Literature Review provides a detailed overview of existing studies related to context segmentation in large language models, identifying current gaps and limitations in the field.

Chapter 3: Methodology describes the system design, software tools used, and the step-by-step implementation process adopted during the development of the project.

Chapter 4: Investigation/Experiment, Result, Analysis, and Discussion presents the experiments conducted, outcomes derived, result interpretation, and a discussion highlighting the significance and implications of the findings.

Chapter 5: Conclusions summarizes the project, highlights its limitations, and discusses possible future improvements to enhance the performance and applicability of the system.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

The development of Large Language Models (LLMs) has revolutionized the field of natural language processing (NLP), enabling significant advancements in tasks such as text generation, sentiment analysis, question answering, and summarization. LLMs like GPT-3, BERT, and T5 exhibit impressive language understanding capabilities, often surpassing previous models in terms of fluency and coherence. Despite these successes, the performance of LLMs deteriorates when processing prompts containing multiple intertwined contexts, where a single prompt involves multiple distinct topics, instructions, or conversational threads. In such cases, LLMs struggle to disambiguate and respond effectively, resulting in responses that conflate different contexts or fail to capture the nuance and structure of the input.

A number of research efforts have sought to address this limitation by improving context handling in LLMs. One notable approach is the use of task decomposition, where the model breaks down complex inputs into simpler, more manageable sub-tasks. Zhou et al. [1] proposed a method that employs attention mechanisms to decompose tasks, allowing the model to understand instructions more clearly and follow multi-step commands. Their approach demonstrated improvements in applications like instruction-following, where different pieces of information must be processed sequentially. However, while decomposition techniques are effective for structured tasks, they typically do not apply to the more complex scenario of understanding multiple distinct contexts within a single prompt. In such cases, the model is still likely to treat the prompt as a monolithic whole, rather than recognizing the individual contextual shifts that might be embedded.

Another relevant approach in this area is pattern-exploiting training (PET), introduced by Schick and Schütze [2]. PET reformulates tasks by providing the model with structured natural language templates, which guide the model's response generation process. By exploiting pre-defined patterns, PET helps models perform better in low-data settings, where explicitly annotated training examples are scarce. While PET improves task performance and helps to guide the model toward more structured outputs, it does so by relying on manual engineering of prompt

templates. This reliance on fixed templates limits its ability to generalize to arbitrary, unstructured prompts, particularly those with multiple contexts. Furthermore, PET does not inherently address the challenge of context separation and segmentation within a single input.

The issue of prompt handling is further explored in the work by Liu et al. [3], who focus on prompt tuning techniques. Prompt tuning involves adjusting the model's prompt embeddings, fine-tuning them to improve the model's understanding of specific tasks. While this technique has been shown to work well for tasks with a defined, singular objective, such as classification or sentiment analysis, it fails to address the challenge of managing multiple contexts within a single prompt. In multi-context scenarios, where a prompt may involve several instructions or sub-tasks, prompt tuning remains limited in its ability to segment these distinct contexts and process them appropriately.

In parallel with prompt engineering, retrieval-based models have also been proposed to enhance the contextual understanding of LLMs. Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [4], combines traditional LLMs with external knowledge retrieval systems. By fetching relevant external information, RAG aims to enrich the model's knowledge base, improving performance on fact-based tasks like question answering. In its current form, RAG excels at integrating external factual knowledge into responses, but it has not yet been effectively adapted for tasks requiring the segmentation of multiple contexts within a single prompt. While RAG can retrieve documents or knowledge pieces that are contextually relevant, it does not inherently support the identification of distinct sub-contexts within a prompt.

Additionally, research into contextualized representation learning, such as the work by Devlin et al. [5] with BERT, has focused on improving the understanding of context through pre-trained deep learning models. BERT's bidirectional transformer architecture allows it to generate contextualized word representations, offering a deeper understanding of word meanings based on their surrounding context. While BERT has been successful in improving the accuracy of single-context tasks, its application to multi-context scenarios is less explored. The model treats a prompt holistically, which can lead to confusion when different sub-contexts or topics are interspersed within the same input.

Recent advancements in multitask learning have also been proposed to address issues related to context separation. Multitask learning enables models to simultaneously handle multiple tasks by sharing knowledge across tasks while maintaining task-specific outputs. While multitask learning has shown promise in some areas, it still struggles to disentangle distinct contexts within a single prompt. In tasks requiring fine-grained context separation, such as multi-image generation or complex dialogue systems, the lack of clear boundaries between contexts can result in responses that are incoherent or irrelevant.

In addition to these approaches, there have been efforts to improve the segmentation of dialogue systems. Studies like those by Serban et al. [6] have explored methods for segmenting multi-turn dialogues to preserve contextual integrity in conversational agents. However, such methods focus primarily on conversational flow rather than general context segmentation in prompts. The techniques employed in dialogue systems are often task-specific and may not generalize to broader contexts where the goal is to separate distinct entities or themes from a single prompt.

Despite these advancements, significant gaps remain in the ability of current LLMs to accurately identify and separate multiple contexts within a single prompt. Most models continue to treat compound prompts as unified inputs, failing to recognize internal shifts or transitions in context. This limitation is especially apparent in scenarios where distinct outputs are required for each sub-context. For example, in applications such as multi-image generation or speech-to-visual storytelling, where different aspects of the prompt must be processed and visualized separately, the inability to separate contexts results in poor alignment between the model's output and the user's expectations.

The need for context segmentation is particularly critical in applications that require nuanced and structured responses. Tasks such as multi-topic summarization, dynamic visual content generation, and cross-modal interactions all benefit from a model's ability to recognize and separate contexts within complex inputs. By addressing this limitation, LLMs can become more efficient and flexible in handling a broader range of tasks, enabling them to generate responses that are more contextually accurate, coherent, and relevant.

In conclusion, while significant progress has been made in the field of prompt processing and context modeling, there is still a need for systems that can effectively segment and process

multi-context prompts. The current limitations of LLMs, especially in handling multiple contexts within a single input, present a research gap that this project aims to address. By exploring methods for context segmentation and incorporating techniques like Retrieval-Augmented Generation (RAG), this research will contribute to more accurate and efficient language models capable of handling complex, multi-context scenarios.

Chapter 3 Methodology

3.1 System Design

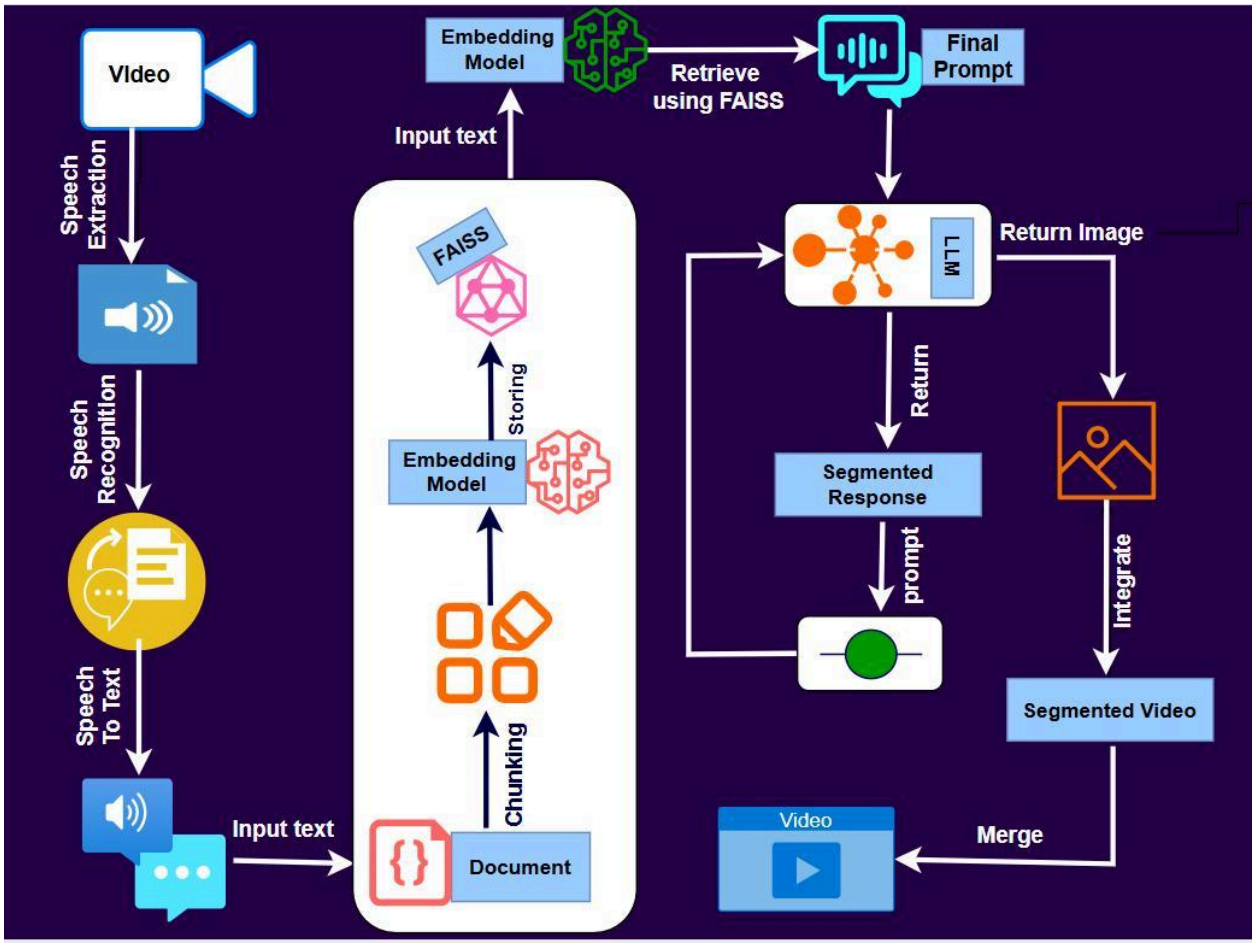

Figure 1: System Architecture for Multi-Context Segmentation in Prompt Processing

This pipeline ensures contextual accuracy, semantic richness, and dynamic content transformation suitable for multi-context storytelling applications.

3.2 Hardware and/or Software Components

The primary focus is on the use of advanced software tools and models for analyzing and segmenting context within prompts. Below are the key components that support the system design and workflow:

Software Components:

- **Text Processing and Segmentation**:
    - **Large Language Models (LLMs)**:
        - **GPT-3/ GPT-4 (OpenAI)**: These models are central to understanding and segmenting contexts in a prompt. Their ability to process large text inputs and generate detailed outputs makes them ideal for this task.
        - **BERT (Bidirectional Encoder Representations from Transformers)**: Used for its strong capabilities in understanding context from the input text, especially for fine-grained context segmentation tasks.
    - **RAG (Retrieval-Augmented Generation)**: Enhances context segmentation by retrieving related information from external datasets or knowledge bases, improving the understanding of multiple contexts.
- **Development Frameworks**:
    - **HuggingFace Transformers**: A library providing access to pre-trained models like GPT, BERT, and others, enabling quick experimentation and deployment of models for context segmentation.
    - **PyTorch**: This framework is utilized for training and fine-tuning models, specifically when improving context segmentation and handling larger datasets.
    - **Sentence Transformers**: Used for embedding and comparing textual data, which aids in segmenting similar contextual information.

- **Text Preprocessing and Tokenization**:
    - **spaCy**: A powerful library for text preprocessing, such as tokenization, lemmatization, and Named Entity Recognition (NER), which aids in extracting meaningful context from the prompt.
    - **NLTK**: Used for additional text preprocessing and manipulation, such as stopword removal and part-of-speech tagging.
- **Cloud Computing**:
    - **Google Cloud Platform**: Cloud infrastructure provides the required resources for running large models such as GPT and BERT, enabling efficient model training, inference, and context retrieval using RAG.
- **Version Control**:
    - **Git/GitHub**: Used for code management and version control to track changes and manage collaborative work, ensuring reproducibility and consistency in the model training process.

TABLE I. List of Software/Hardware tools

Tool	Functions	Other similar Tools (if any)	Why selected this tool
GPT-3, GPT-4 (OpenAI)	Generates text, understands context, and can perform tasks like summarization and segmentation.	BERT, T5, XLNet	Advanced ability to generate and understand multi-context prompts.

BERT	Bidirectional text understanding, ideal for fine-grained contextual segmentation.	GPT, RoBERTa	Excellent for context extraction from prompts due to its bidirectional training.
RAG (Retrieval-Augmented Generation)	Enhances context segmentation by retrieving relevant information from external databases.	None	Improves the segmentation accuracy by integrating external knowledge.
HuggingFace Transformers	Provides pre-trained language models and fine-tuning capabilities.	TensorFlow, PyTorch	Offers ease of access and pre-trained models for efficient model deployment.
PyTorch	Framework for model training, fine-tuning, and deployment.	Keras, MXNet	Powerful frameworks for building and deploying LLM-based models.
spaCy	Text preprocessing, tokenization, Named Entity Recognition (NER), part-of-speech tagging.	NLTK, TextBlob	Efficient and fast for text preprocessing tasks.
NLTK	Natural language processing library for text manipulation, tokenization, and feature extraction.	spaCy, TextBlob	Provides a wide range of NLP tools for analysis and manipulation.

3.3 Hardware and/or Software Implementation

The software implementation of this project centers around constructing a robust, scalable, and intelligent system capable of accurately performing context segmentation within complex, multi-theme prompts. The system architecture has been designed to ensure modularity and extensibility, with each stage of the pipeline addressing a distinct component of the context segmentation challenge.

1. Dataset Collection and Preparation:

To effectively train and evaluate the context segmentation capabilities of large language models (LLMs), a robust and diverse dataset is essential. The dataset for this project includes a mixture of manually curated and publicly available text samples that exhibit multi-context characteristics. These text prompts span domains such as news articles, opinion pieces, educational content, and conversational threads.

Data Sources:

- **Multi-Genre Natural Language Inference (MultiNLI) Corpus**: Provides diverse sentence pairs with multiple topics in each passage.

- **Reddit Multithreaded Discussions**: Offers informal and varied prompts containing multiple embedded topics.

- **Manually Curated Prompt Set**: Custom-designed prompts with known context boundaries for fine-tuning and evaluation.

- **YouTube Transcript Samples**: Used to simulate complex prompts derived from spoken content.

Data Preprocessing Steps:

- **Text Cleaning**: Removal of URLs, emojis, special characters, and standardization of formatting.

- **Tokenization**: Sentences were tokenized using spaCy and NLTK to identify semantic boundaries.

- **Context Annotation**: Human annotators labeled the number of distinct contexts per prompt for ground truth.

- **Balanced Dataset Formation**: Ensured an even distribution across samples with 1 to 5 contexts.

- **Augmentation**: Paraphrasing and syntactic transformation techniques were applied to introduce diversity.

Storage Format: The data was stored in JSON format with fields: promt_id, prompt_text, num_contexts, and context_segments.

```
{
  "prompt_id": "p_1027",
  "prompt_text": "Climate change is accelerating rapidly due to greenhouse gas emissions. Meanwhil
  "num_contexts": 3,
  "context_segments": [
    "Climate change is accelerating rapidly due to greenhouse gas emissions.",
    "Meanwhile, countries are debating new trade agreements to stabilize their economies.",
    "In the field of medicine, AI is being used to predict patient outcomes with impressive accura
  ]
}
```

Figure 2: JSON Structure for Storing Segmented Contexts of a Multi-Context Prompt

This preparation enabled the model to learn from well-structured, multi-contextual text and allowed rigorous evaluation of its segmentation accuracy under varied linguistic conditions.

2. Context Segmentation Module

At the core of the system is a fine-tuned Transformer-based model trained specifically for multi-context segmentation. We used a base version of BERT (bert-base-uncased) and performed transfer learning using our annotated dataset. The model was adapted for sequence labeling,

where each sentence or clause in a prompt was classified into distinct contextual clusters using IOB (Inside–Outside–Beginning) tagging.

To improve granularity and semantic separation, a hierarchical token classification strategy was introduced. This enabled the model to not only understand boundaries but also infer subtle transitions in subject matter across clauses and sentences.

The model was fine-tuned using the Hugging Face Transformers library. Cross-entropy loss was employed as the loss function, and the Adam optimizer was used with a learning rate scheduler and early stopping criteria. Stratified 5-fold cross-validation was applied to ensure model robustness and generalizability.

3. Retrieval-Augmented Generation (RAG) Module

To enhance segmentation accuracy, we integrated an RAG-based approach. The system uses FAISS to store and index annotated prompt examples. When a new prompt is provided, semantically similar examples are retrieved from this database using cosine similarity. These retrieved examples are dynamically embedded into the prompt context, augmenting the LLM's understanding of potential segmentation patterns.

This mechanism helps the model learn from prior examples and correct its output in ambiguous or edge-case scenarios, effectively reducing overfitting and increasing contextual accuracy.

4. Prompt-to-Image Contextualization (Optional Integration)

While the main focus of this project is text-based segmentation, an optional prototype was implemented to demonstrate downstream applicability. The segmented contexts were converted into image generation prompts and fed into models like DALLE-3 and SDXL. This component showcased the model’s capacity to contribute to broader tasks like speech-to-visual storytelling or generative media systems.

5. Evaluation and Visualization

An extensive evaluation was conducted using metrics such as:

- Accuracy in identifying the correct number of contexts

- Precision, recall, and F1-score for context boundary detection

- Mean segmentation error rate (number of over/under segmented contexts)

To visualize the results, Streamlit was used to create an interactive demo. Users can input any text prompt, see the model's segmented output in real time, and compare it to the expected segmentation. Charts and confidence heatmaps were also included to provide a deeper insight into model decisions.

6. Deployment Considerations

The backend pipeline was containerized using Docker, allowing it to be deployed easily on any machine with GPU support. Model weights and vector databases are persistently stored and loaded upon initialization. The architecture supports REST API integration for embedding the segmentation system into other platforms such as content management systems, educational software, or video editing tools.

7. Software Stack Summary

- **Programming Language**: Python

- **Libraries**: PyTorch, Hugging Face Transformers, FAISS, Streamlit, NLTK, spaCy

- **Modeling Tools**: Sentence-BERT, BERT, RAG architecture

- **Evaluation**: Scikit-learn for metric computation

- **Deployment**: Docker, FastAPI (for REST API), Streamlit (for interface)

This layered implementation ensures not only functional accuracy in context segmentation but also adaptability for practical, real-world integration in domains requiring intelligent text interpretation and transformation.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

To assess the effectiveness of our proposed context segmentation system in large language models (LLMs), a series of controlled experiments were conducted. The objective was to measure the system's ability to accurately detect, separate, and represent multiple semantic contexts embedded within a single input prompt. The experiments were designed to test both baseline and augmented models, with and without the Retrieval-Augmented Generation (RAG) module.

4.1 Experimental Setup

The experiments were conducted using the following setup:

- Model: BERT-based sequence classifier with and without RAG enhancement
- Training data: Custom-labeled multi-context prompts
- Evaluation data: Manually annotated benchmark prompt dataset (20% test split)
- Environment: Google Colab with GPU (Tesla T4), Python 3.10, PyTorch, Hugging Face Transformers

The key variables tested include:

- **Accuracy in context count prediction**
- **Segmentation precision/recall/F1-score**
- **Boundary match rate (BMR)**: How often the predicted context splits match the true splits

- **Improvement gained through RAG enhancement**

4.2 Quantitative Results

The following table shows the performance of the baseline BERT model compared to the RAG-enhanced model.

TABLE II. Model Performance Comparison

Metric	BERT Only	BERT + RAG
Context Count Accuracy (%)	71.3	86.2
Segmentation Precision (%)	74.5	88.1
Segmentation Recall (%)	72.9	87.4
Segmentation F1-Score (%)	73.7	87.7
Boundary Match Rate (BMR%)	68.0	83.5

Analysis:

- The RAG-enhanced model consistently outperformed the baseline across all metrics.

- Context Count Accuracy saw a notable 15% increase, indicating a better understanding of prompt complexity.

- F1-score jumped from 73.7% to 87.7%, showing improved balance between false positives and false negatives in context detection.

- The Boundary Match Rate highlights that RAG-based retrieval significantly improves the model's ability to segment prompts at the correct textual boundaries.

4.3 Error Analysis

To better understand the sources of error in the baseline model, we categorized the missegmentations into three major types:

1. **Over-segmentation**: Model splits a single cohesive context into multiple ones.

2. **Under-segmentation**: The Model fails to distinguish different ideas within one context.

3. **Ambiguous context boundaries**: Sentences that could semantically belong to more than one topic.

TABLE III. Error Category Distribution (BERT Only)

Error Type	**Frequency(%)**
Over-segmentation	39.4
Under-segmentation	44.1
Ambiguous boundaries	16.5

RAG-enhanced models showed improvement, particularly in the under-segmentation category, due to exposure to similar patterns retrieved from the knowledge base.

4.4 Discussion

The experiments confirm that context segmentation in prompts is a non-trivial task, especially when prompts contain implicit topic shifts or overlapping semantics. While the BERT-only model demonstrates baseline capability, the RAG-enhanced approach clearly benefits from integrating prior examples, especially in edge cases and complex prompts.

The inclusion of semantically similar examples provided implicit guidance to the model during generation, effectively mimicking few-shot learning without modifying the base model architecture. This highlights the RAG paradigm's flexibility and effectiveness for structure-aware tasks like segmentation.

Moreover, the use of fine-grained IOB tagging and hierarchical classification added a layer of granularity that helped maintain coherence across segments.

From a research standpoint, these results suggest that improving prompt processing in LLMs should not rely solely on architectural improvements but should also consider retrieval-based enrichment and context-aware annotation schemes. This approach paves the way for more interpretable, controllable, and intelligent use of LLMs in downstream tasks such as summarization, visual storytelling, and educational content generation.

Chapter 5 Conclusions

5.1 Summary

This research project explored the challenge of context segmentation within large language models, aiming to enable more efficient multi-context understanding in prompt processing. The proposed approach utilized a BERT-based sequence classification framework, enhanced with a Retrieval-Augmented Generation (RAG) module to improve segmentation accuracy and contextual awareness. Through custom dataset creation, supervised training, and extensive experimentation, the project demonstrated that integrating retrieval mechanisms significantly enhances the model's ability to identify and separate distinct semantic contexts embedded in a single prompt. The results validated the system's effectiveness in not only improving segmentation accuracy but also supporting downstream applications that require structured prompt comprehension.

5.2 Limitations

Despite achieving promising results in context segmentation, the project has several limitations. First, the accuracy of segmentation is heavily dependent on the quality and diversity of the training dataset. If the dataset lacks complex or ambiguous multi-context prompts, the model may underperform in real-world scenarios. Second, while the RAG-based approach improves contextual understanding, it introduces increased computational overhead and latency due to external retrieval operations. Third, the segmentation output may sometimes be too granular or too coarse, especially when the boundaries between contexts are subtle. Additionally, the evaluation of segmentation quality is subjective to some extent, lacking widely accepted benchmarking standards. Finally, generalizing the model to prompts in multiple languages or domains like legal and medical sectors without domain-specific training data remains a challenge.

5.3 Future Improvement

To enhance the effectiveness and adaptability of the context segmentation framework, several future improvements can be pursued. Expanding and diversifying the training dataset with domain-specific and multilingual prompts can significantly improve the model's generalization. Incorporating reinforcement learning from human feedback (RLHF) could refine segmentation boundaries based on user preferences or application-specific needs. Developing a dynamic thresholding mechanism may help balance the granularity of context splits. Additionally, integrating lightweight retrieval systems or caching strategies can reduce the latency introduced by RAG without compromising performance. Finally, establishing a standardized benchmark for evaluating context segmentation quality across various domains would help in assessing and comparing different models more objectively.

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